SynthAgent: Adapting Web Agents with Synthetic Supervision

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Paper version: arXiv 2511.06101v3 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Web agents struggle on new websites: task demonstrations are scarce.*

Synthetic tasks **hallucinate** and trajectories are **noisy**. The paper uses no human-written tasks.

Motivation: a task-grounding tension

1. **Hallucinated tasks.** One-step proposals name absent items.
2. **Wandering edits.** Explorer: 8.6 per task, 68.3% over budget.
3. **Noisy traces.** Task edits misalign early steps.

Evaluation setup

WebArena · 226 tasks | Online-Mind2Web · 300 | Qwen2.5-VL-7B | UI-TARS-1.5-7B | GPT-4.1 synthesizer | 30-step budget | seeds not reported

- **Success rates** from runs; the paper calls WebArena offline, Online-Mind2Web online.
- **Held out:** OM2W sites only; WebArena sites are explored.

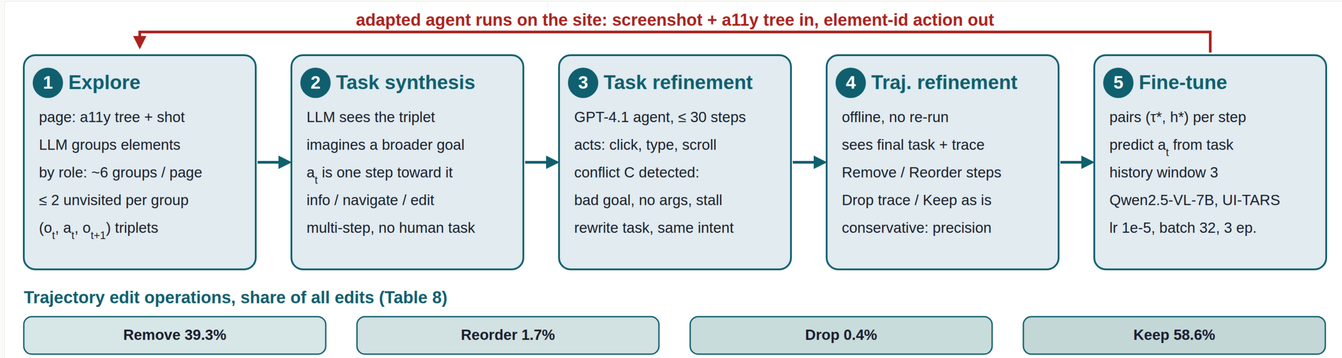
Our read (CUA Radar)

71	★★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out OM2W; matched baselines; ablation; code.
- **Not reported:** seeds; judge check. **AP5:** explored sites are WebArena test sites.
- Pattern K14. Authors' numbers, not rerun.

Method: refine the task, then the trajectory, then SFT

- **Sees / acts:** URL, accessibility tree and screenshot in; click, type, hover, press, scroll, goto by element id out.
- **New:** categorized exploration, conflict-triggered task refinement, offline trajectory refinement.
- **Signal:** no human labels or test tasks; GPT-4.1 writes and edits; plain SFT with a 3-step history.



Our redraw (not the paper's figure): four synthesis stages, then fine-tuning; edit shares from Table 8.

$$C(h_t, \tau_t) = \neg\,\mathrm{ExistsUI}(h_t, \tau_t) \;\vee\; \mathrm{MissingArgs}(h_t, \tau_t) \;\vee\; \mathrm{Stall}(h_t)$$

the task is rewritten only when a conflict is detected

Key results: WebArena success rate (%)

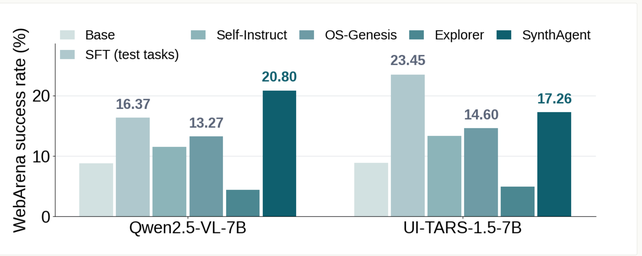


Table 1, redrawn: 226 tasks, one reported value per cell, no std; the paper calls WebArena offline.

Above other synthetic methods; UI-TARS stays below test-task SFT

Ablation: WebArena success rate (%)

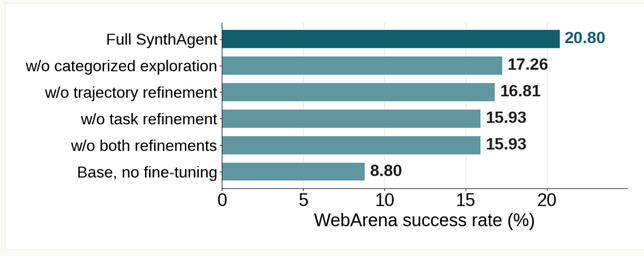


Table 10, redrawn: one component removed at a time (rows match the Qwen2.5-VL-7B rows of Table 1).

SO WHAT → *Task refinement costs most (−4.87), then trajectory refinement (−3.99) and exploration (−3.54). Removing both refinements gives the same 15.93.*

Headline Numbers

20.80%
WebArena, Qwen2.5-VL-7B
vs 8.80% base

24.34% Qwen3-VL-8B
23.56% OM2W judge avg
+10.2 avg vs base

The paper's own controls

CONTROL	OURS	COMPARED WITH
OM2W GPT-4.1 judge	31.67	OS-Gen 19.53
OM2W GPT-5.1 judge	15.67	OS-Gen 11.00
Qwen3 OM2W GPT-5.1	15.00	base 14.67
Done / over budget	96.5 / 0.0%	Explorer 30.5 / 68.3%

LLM judges; no human check

Takeaway

Refinement lifts 7B agents on WebArena; Online-Mind2Web gains are smaller and judge-dependent.

GPT-4.1 synthesizes + refines

API cost per trajectory: 0.13 vs 0.22 USD

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